

Case 001: Read-Level Diagnosis of Time-Variable Contamination in a HgCdTe Focal Plane Array

Independent analysis of public Hubble WFC3/IR MULTIACCUM measurements

Engineering question: Which nondestructive-read intervals are contaminated, how long does the contamination persist after the obvious spatial gradient disappears, and how does read selection change the reconstructed count-rate product?

Metric	Result
Public products	2 IMA ramps + 2 FLT products
Positive-read intervals	14
Spatially anomalous intervals	0, 1, 2, 3
Common-mode intervals at 0.05 e-/s/pixel	0, 1, 2, 3, 4, 5, 6
Peak transient excess	0.4958 e-/s/pixel
Integrated positive excess	197.26 e-/pixel
Excess / nominal integrated charge	16.10%

Executive finding

The exposure contains two anomaly regimes. The earliest reads exhibit a strong left-side enhancement, while a lower-amplitude detector-wide excess remains after the spatial distortion is no longer obvious. A spatial-uniformity check alone therefore understates the contamination duration.

Matched-control subtraction, source-free masking, threshold sensitivity, spatial block bootstrap, and independent ramp reconstruction support a time-variable external background. They do not support an intrinsic HgCdTe material defect.

Public-data demonstration; no client data are represented.

1. Read-level method and spatial evolution

For each calibrated read, cumulative charge is reconstructed as $Q_i = R_i t_i$. Consecutive-read rate is then $(Q_i - Q_{i-1})/(t_i - t_{i-1})$. Permanent detector-quality bits are rejected. DATAREJECT=8192 is retained as a diagnostic because it is generated by the up-the-ramp algorithm and can respond to the time-variable background itself.

Representative WFC3/IR instantaneous interval-rate maps

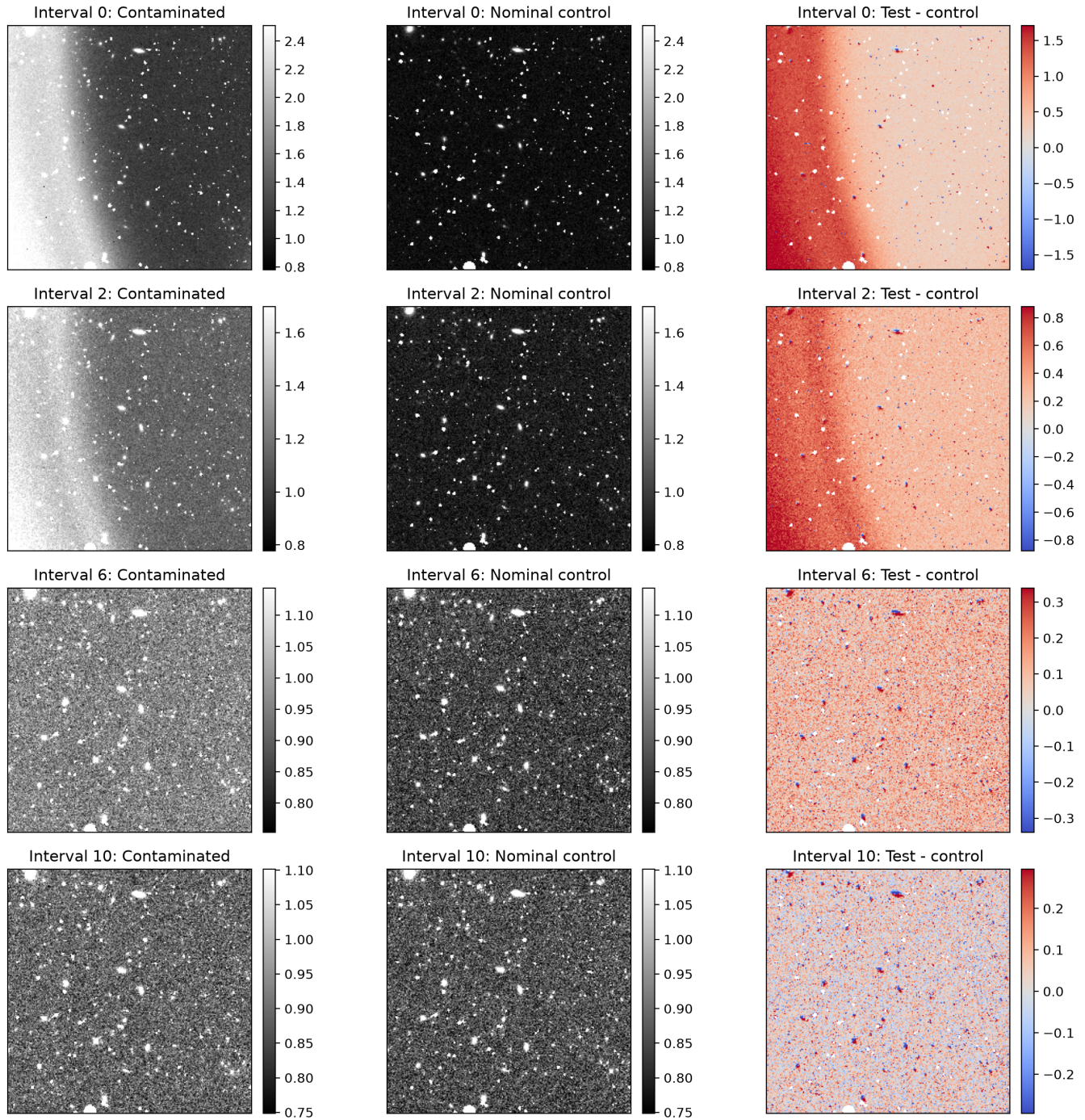


Figure 1. Contaminated exposure, nominal control, and matched difference for representative intervals. Early spatial enhancement transitions into a weaker common-mode tail.

2. Control-relative transient diagnosis

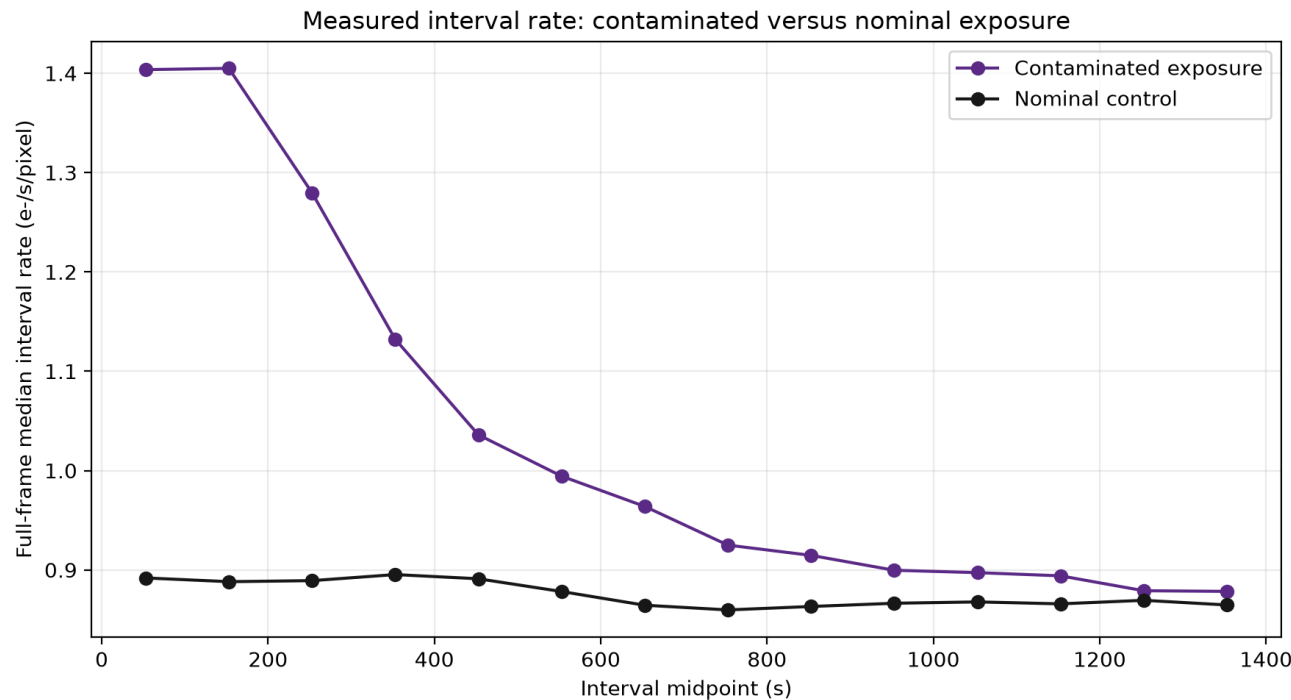


Figure 2. Full-frame interval rate for the affected and nominal exposures.

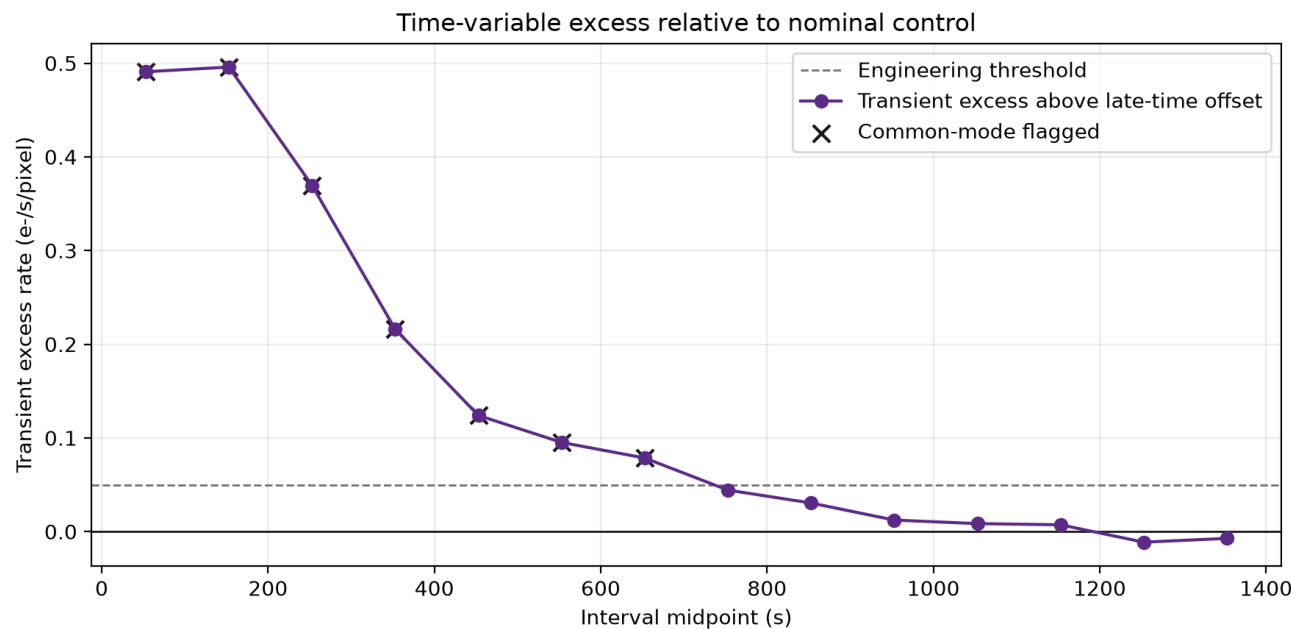


Figure 3. Late-offset-corrected transient excess. Nominal common-mode classification: intervals 0, 1, 2, 3, 4, 5, 6.

Metric	Result
Common-mode duration	700.0 s
Late-time static offset	0.02096 e-/s/pixel
Peak asymmetry: affected / control	53.85% / 1.83%
Pixels ever carrying DATAREJECT	35.80%

3. Robustness and uncertainty

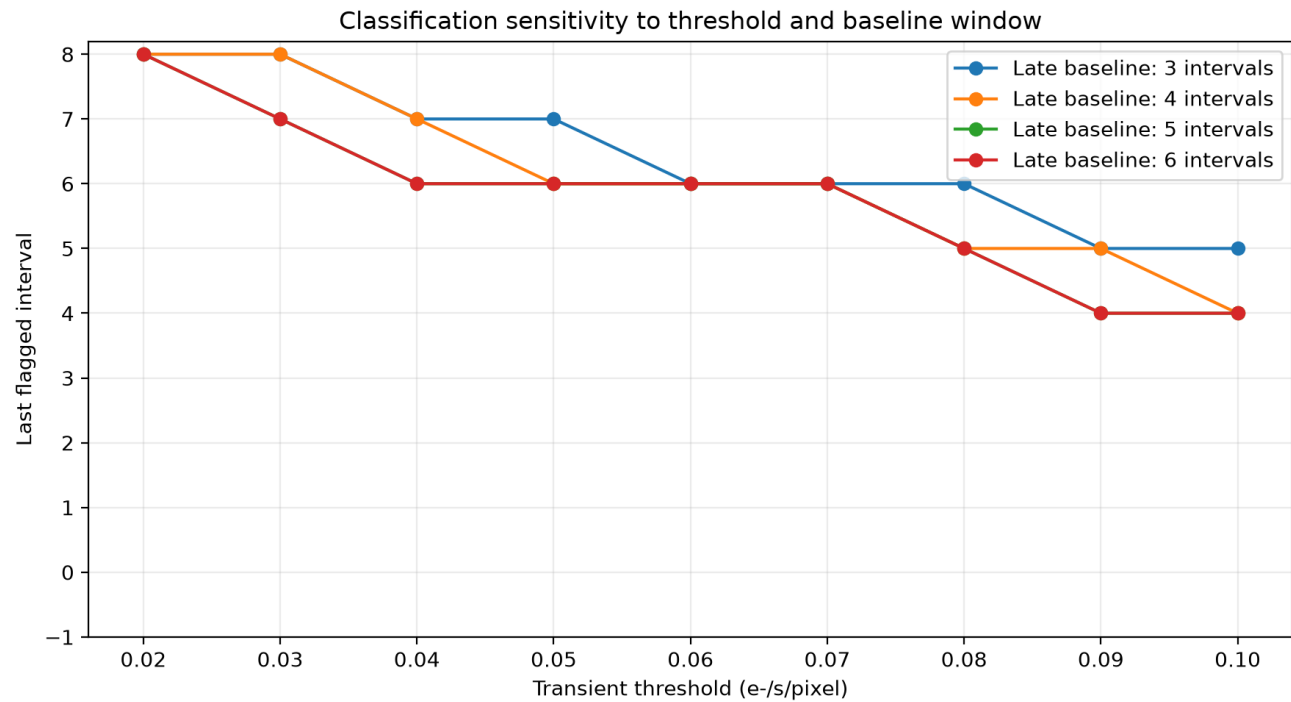


Figure 4. Across thresholds 0.02-0.10 e-/s/pixel and late baselines of 3-6 intervals, the last flagged interval ranges from 4 to 8.

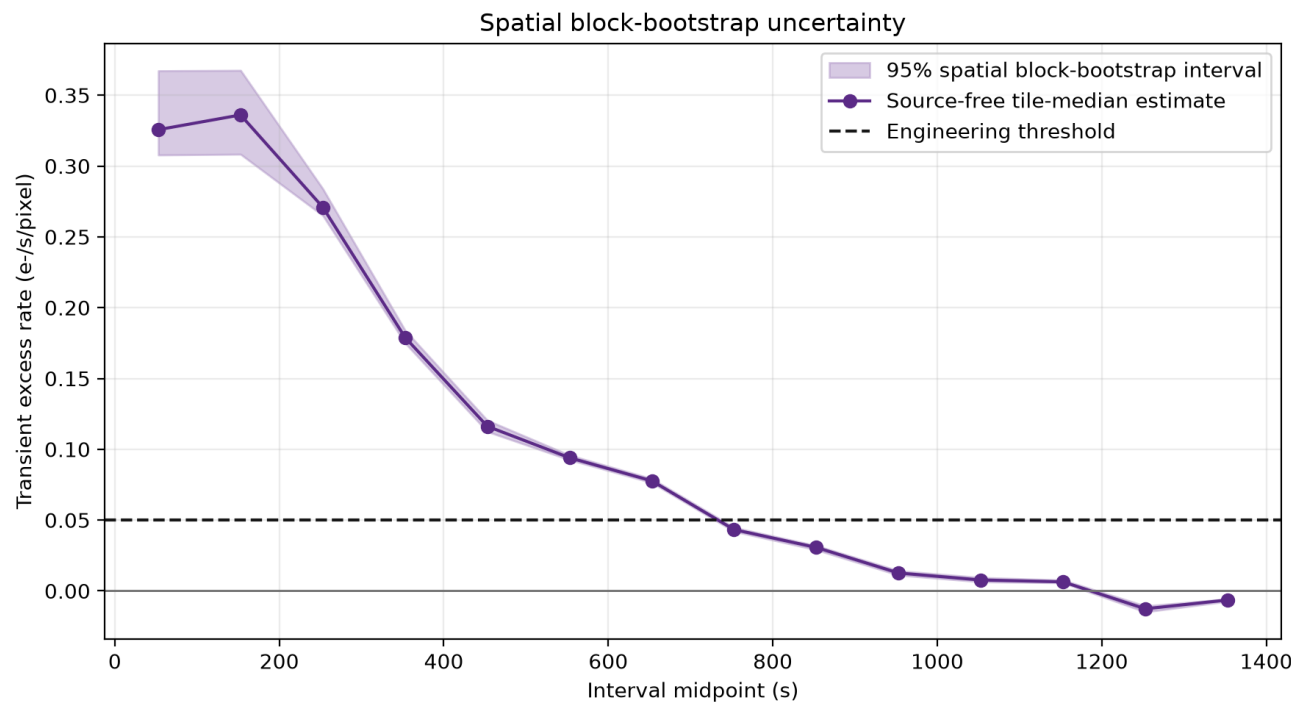


Figure 5. 95% spatial block-bootstrap intervals using 32 x 32 pixel tiles.

Metric	Result
Source-free pixels retained	94.25%
Source-free classification agreement	100.0%
Bootstrap tiles / replicates	779 / 2000
Integrated excess, 95% interval	144.61 to 159.50 e-/pixel
Flagged duration, 95% interval	700.0 to 700.0 s

4. Independent ramp reconstruction

A free-intercept, DQ-aware ordinary-least-squares slope is fitted to cumulative charge. The post-transient fit begins after the final nominal common-mode interval, so accumulated early charge changes the intercept rather than forcing the later slope upward.

Archived and independently reconstructed count-rate products

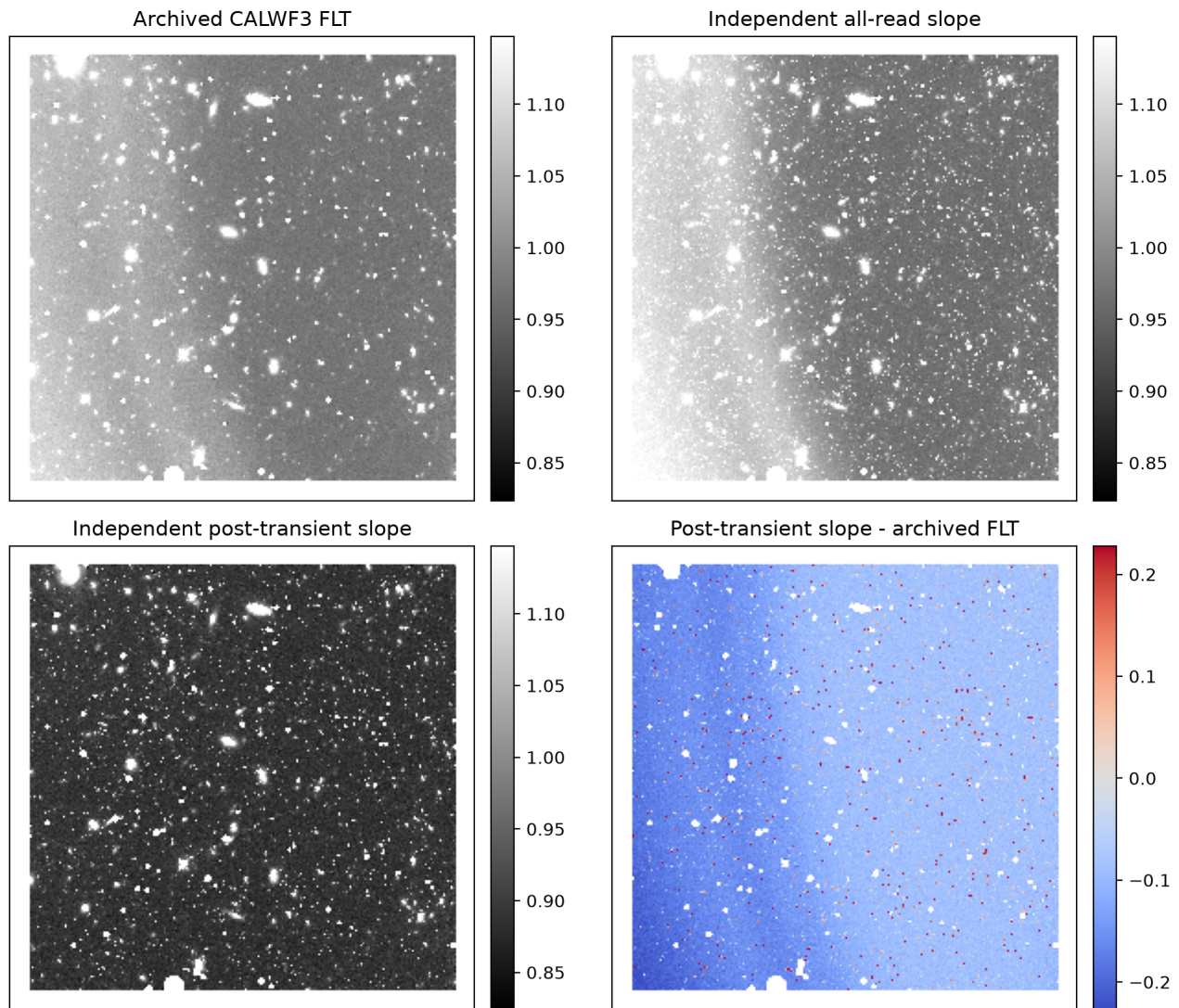


Figure 6. Archived CALWF3 FLT, independent all-read slope, independent post-transient slope, and post-transient-minus-FLT residual.

Metric	Result
Post-transient fit begins	$t = 702.934 \text{ s}$
Archived FLT source-free median	1.00296 e-/s/pixel
Independent all-read median	0.99992 e-/s/pixel
Independent post-transient median	0.89337 e-/s/pixel
Post-transient vs FLT residual NMAD	0.05243 e-/s/pixel

5. Engineering conclusion

The anomaly is most consistent with a time-variable external background: it decays smoothly, begins with a spatial gradient, develops a common-mode tail, is absent from the matched control, and changes the slope inferred from the nondestructive ramp.

Metric	Result
Time-variable external background	Most consistent
Stable gain or flat-field nonuniformity	Disfavored
Intrinsic dark-current nonuniformity	Strongly disfavored
Readout-quadrant offset	Secondary contribution possible
Isolated cosmic ray	Cannot explain detector-wide smooth decay

Recommended disposition: inspect consecutive-read maps before accepting a final slope product; classify spatial and common-mode effects separately; use a matched control or stable late-ramp baseline; quantify threshold sensitivity; and refit after excluding the transient regime. Do not infer detector material defects until measurement-condition explanations have been rejected.

Interpretation boundary: this case does not infer HgCdTe composition, carrier lifetime, dark-current mechanism, detectivity, or intrinsic material quality.

Primary references: STScI WFC3/IR IMA Visualization Tools; WFC3 Data Handbook Sections 2.2, 3.3, and 7.10; STScI scattered-light correction notebook.

Brooks Photonics | Independent physics-based infrared detector analysis | brooks-photonics.com | terence@brooks-photonics.com